

MOHAMED ELAZAB

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Experience

Research Electronics Engineer

Jan 2024 – Present

The MetroHealth System – MetroHealth Center for Rehabilitation Research

Cleveland, OH

- Configure hardware and software to investigate the application of electrical nerve block in animal experiments
- Perform electrochemical testing and determine the reversible charge injection limits of nerve blocking electrodes
- Develop LabVIEW programs for data acquisition and control of lab instruments and analyze data in MATLAB
- Assist with writing grant proposals, manuscript publications and slides for conference talks

Graduate Research Assistant

Aug 2021 – Dec 2023

Cleveland Clinic – Lerner Research Institute

Cleveland, OH

- Conducted device-development research at the Urological Biomechanics lab
- Collected, interpreted, analyzed and compiled *in-vivo* wireless bladder sensor data in preclinical and clinical trials
- Developed LabVIEW and MATLAB programs to automate wireless sensor data collection and analysis in miscellaneous lab projects
- Presented lab research at 3 conferences: the 2022 Experimental Biology meeting and the 2022 & 2023 Society for Pelvic Research meetings

Biomedical Engineer 1

Sep 2020 – Jun 2021

Inova Health System – Inova Loudoun Hospital

Leesburg, VA

- Performed preventive maintenance, calibration and repair of diagnostic and therapeutic medical devices at Inova Loudoun Hospital and Inova primary care groups in Loudoun County, VA
- Provided technical support to hospital staff needing assistance with medical hardware and software

Education

Master of Science in Computer Science

Jan 2025 – Present

Georgia Institute of Technology

Atlanta, GA

Coursework: Operating Systems, Artificial Intelligence

Master of Science in Biomedical Engineering

Aug 2021 – May 2024

Cleveland State University

Cleveland, OH

Coursework: Embedded Systems, Software Defined Radio, Biomedical Signal Processing, Biomedical Instrumentation, Medical Devices, Python with Mechanical Engineering Applications

Bachelor of Science in Bioengineering

Aug 2016 – May 2020

Temple University

Philadelphia, PA

Dean's List: Spring 2018, Fall 2019, Spring 2020

Projects

Computational modeling of dorsal and ventral spinal cord stimulation for pain neuromodulation

- Built an open-source finite element pipeline (FreeCAD → Gmsh → ElmerFEM) on the RADO-SCS spine model, with editable lead placement, to compare electric fields from dorsal and ventral epidural leads at the dorsal columns and spinothalamic tracts
- Developed a Python Lead Designer tool to configure dorsal, ventral, and DRG leads, check each lead's fit in the epidural space and clearance from neighboring leads, and export STL geometry for meshing; presented as a poster at MetroHealth Research Day 2026

Embedded bioimpedance measurement system using the AD5940

- Developed bare-metal C firmware for the AD5940 bioimpedance front-end on an ADuCM3029 (Cortex-M3) to run 2-wire and 4-wire impedance sweeps (1–200 kHz) and continuous single-frequency time-series measurements, using startup code and linker scripts with an `arm-none-eabi-gcc` toolchain
- Built a Python GUI to flash firmware, live-plot impedance and phase over UART, and record data to CSV; replaced text output with a checksummed binary frame protocol to fix heap-fragmentation hangs during long continuous runs

Remotely-controlled closed-loop laser controller for *in-vivo* photobiomodulation therapy

- Developed LabVIEW software for outputting configurable waveforms from the LDC-3736 precision laser controller and for measuring laser power from the Ophir NOVA II power meter using LabVIEW's NI-VISA API

Embedded C software for Texas Instruments DAC63004W evaluation module

- Prototyped a biphasic current waveform source for an awarded NIH research grant

Electrochemical evaluation of carbon slurry electrodes using pyVISA

- Automate the charge injection capacity testing of custom-made carbon slurry electrodes

Skills

- **Software:** MATLAB, LabVIEW, Arduino, Git, GNU/Linux command line tools, FreeCAD, Gmsh, ElmerFEM
- **Lab Equipment:** Digital Multimeter, Waveform Generator, Power Supply, Oscilloscope, Source Measure Unit, Potentiostat, National Instruments data acquisition systems, Formlabs 3D Printer
- **Programming Languages:** C, C++, Python, Scheme, Bash

Awards

\$500 travel award to attend the 8th annual meeting of the Society for Pelvic Research *December 2023*

- Abstract title: “Bladder Volume Estimation Using a Catheter-free Wireless Pressure-Volume Sensor in Minipigs”

Top 5 abstracts at the 7th annual meeting of the Society for Pelvic Research *December 2022*

- Abstract title: “Catheter-free Bladder Volume Monitoring Using Intravesical Sensor”

Abstracts & Publications

- **Elazab, M.**, Bander, S. A., Kim C. H., Green, D. B., Vrabec, T. L., Van Acker, G. M. (2026, October 2). Computational modeling of dorsal and ventral spinal cord stimulation for pain neuromodulation. The 27th Annual MetroHealth Research Day, Cleveland, OH, United States.
- Bender, S. A., Green, D. B., Thakkar, V. S., Zimmerman, H. L., **Elazab, M.**, Kilgore, K. L., & Vrabec, T. L. (2026). Single-electrode, bidirectional control of heart rate via vagus nerve modulation in rat model. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*. <https://doi.org/10.1109/TNSRE.2026.3663907>
- Green, D. B., Bender, S. A., **Elazab, M.**, Thakkar, V. S., Zimmerman, H. L., Van Acker, G. M., Vrabec, T. L. (2025). Electrophysiological methods to assess peripheral pain block in an anesthetized rat. *J. Vis. Exp.* (225), e68740. <https://doi.org/10.3791/68740>
- Gross, M. D., Frainey, B. T., Lyon, M. E., Lewis, K. C., **Elazab, M.**, Bukhari, H. A., Tevis, T., Butler, R. S., Burhanna, R., Smith, C., Goldman, H. B., Hoang-Roberts, L., Damaser, M. S., & Majerus, S. J. A. (2025). Validation of a wireless catheter-free ambulatory urodynamics device in women with neurogenic bladder. *Neurourology and Urodynamics*. <https://doi.org/10.1002/nau.70172>
- Bender, S. A., Green, D. B., Thakkar, V. S., **Elazab, M.**, Zimmerman, H. L., Vrabec, T. L. (2025). Closed-loop control of transcutaneous direct current nerve block can produce long-lasting partial nerve conduction. *IEEE EMBS 12th Annual International Conference on Neural Engineering*.
- Zareen, F., **Elazab, M.**, Hanzlicek, B., Doelman, A., Bourbeau, D., Majerus, S., Damaser, M., Karam, R. (2024). Optimization of activity-driven event detection for long term ambulatory urodynamics. *Journal of Engineering in Medicine*, 238(6). <https://doi.org/10.1177/09544119241264304>
- Cornier Martinez, S. M., Agrawal, S., Lewis, K., **Elazab, M.**, Lyon, M., Nagle, T., ... Damaser, M. (2024). MP03-13 Artificial urethral sphincter pressure regulating balloon and cuff variations following device explantation. *Journal of Urology*, 211(5S), e27. <https://doi.org/10.1097/01.JU.0001009488.55564.85.13>
- Gross, M. D., Lyon, M., Lewis, K., **Elazab, M.**, Tevis, T., Majerus, S., ... Damaser, M. S. (2024). PD07-02 Clinical testing of a novel wireless ambulatory urodynamic system in patients with neurogenic bladder. *Journal of Urology*, 211(5S), e167. <https://doi.org/10.1097/01.JU.0001008552.16893.70.02>
- **Elazab, M.**, Hanzlicek, B., Doelman, A., Majerus, S., Bourbeau, D., Damaser, M. (2023, December 7–9). Bladder volume estimation using a wireless catheter-free wireless pressure-volume sensor in minipigs. The 8th Annual Meeting of the Society for Pelvic Research, Savannah, GA, United States.
- **Elazab, M.**, Nemer, G., Majerus, S., Damaser, M. (2023, November 29–30). Catheter-free ambulatory urodynamic monitoring of bladder volume. The 14th biennial Incontinence: The Engineering Challenge, Bristol Watershed, Bristol, United Kingdom.
- **Elazab, M.**, Nemer, G., Majerus, S., Damaser, M. (2023, November 15). Towards catheter-free ambulatory monitoring of bladder volume. The 43rd Annual Cleveland Clinic Research Day, Cleveland, OH, United States.
- Majerus, S., Hanzlicek, B., Cabal, D., **Elazab, M.**, Damaser, M., & Bourbeau, D. (2023). PD23-06 Feline ambulatory bladder pressure and volume transmitted by a catheter-free wireless intravesical monitor. *Journal of Urology*, 209(Supplement 4), e672. <https://doi.org/10.1097/JU.0000000000003296.06>
- **Elazab, M.**, Hanzlicek, B., Majerus, S., Bourbeau, D., Damaser, M. (2022, December 8–10). Catheter-free bladder volume monitoring using intravesical sensor. The 7th Annual Meeting of the Society for Pelvic Research, Charlotte, NC, United States.

References

Available upon request.